

HXC-108_liveness_verification

HXC-108 — §11.4.107 Liveness / Freeze-Oracle Verification of Committed Durable Recordings

Date: 2026-06-23 **Scope:** Confirm the committed durable docs/qa/HXC-108_* mp4 recordings show GENUINELY ADVANCING frames (not a frozen/stale single image), per Constitution §11.4.107 (liveness battery / freeze-detection oracle). **Tooling:** ffprobe + ffmpeg (/opt/homebrew/bin, host). **Read-only on recordings.** No commit performed.

Method (§11.4.107 freeze oracle, multi-signal — §11.4.6 honest judgment)

A single captured frame is not proof (§11.4.107). For each mp4 we captured THREE independent signals and combined them rather than trusting any one in isolation:

1. **Frame count + duration + fps** (ffprobe -count_frames) — a 1-frame / single-image video FAILs liveness.
2. **Distinct-frame count** via mpdecimate (drops near-duplicate frames; survivors = genuinely distinct content frames) — this is the perceptual freeze oracle. **1 distinct frame ⇒ frozen-stale.**
3. **First-frame vs last-frame distinctness** (frame-accurate PNG extract + md5) — first≠last ⇒ content genuinely changed across the window (advancing), not one stale image.
4. **freezedetect=n=-60dB:d=0.5** captured for reference. NOTE (§11.4.6 honest boundary): the default -60 dB threshold is extremely sensitive and flags a legitimately-static TUI/status screen as “frozen”; freezedetect ALONE cannot distinguish *rendered-but-static real UI* from *frozen-stale-bluff*, so it is used only as a corroborating signal — the distinct-frame count + first≠last are the decisive oracles.

Verdict vocabulary

- **live** — many distinct frames advancing across the window (genuinely-advancing).
 - **static-real** — real genuine content, few changes (a status/list screen that renders then sits); ≥2 distinct frames AND first≠last ⇒ NOT frozen, acceptable per §11.4.107.
 - **FROZEN-BLUFF** — exactly 1 distinct frame the whole duration / first==last ⇒ real defect, re-record.
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Per-mp4 results (§11.4.118 enumerated coverage — 12 of 12 committed mp4s checked)

#	Recording	Frames	Dur (s)	fps	Distinct (mpdecimate)	first≠last	freezedetect	
1	HXC-108_android/helixcode-android-client.mp4	33	19.92	~1.7 avg	25 (76%)	DIFFERENT	minimal	
2	HXC-108_cli/helixcode-cli-command_exec-...mp4	33	3.30	10	4 (12%)	DIFFERENT	partial	
3	HXC-108_cli/helixcode-cli-generate-...mp4	68	6.80	10	7 (10%)	DIFFERENT	partial	
4	HXC-108_cli/helixcode-cli-version-...mp4	35	3.50	10	5 (14%)	DIFFERENT	partial	
5	HXC-108_tui_server/helixcode-server-api-...mp4	48	4.80	10	5 (10%)	DIFFERENT	partial	
6	HXC-108_tui_server/helixcode-tui-dashboard-...mp4	131	13.10	10	6 (5%)	DIFFERENT	periodic	liv @0,0
7	HXC-108_tui_views/helixcode-tui-projects-...mp4	50	5.00	10	3 (6%)	DIFFERENT	mostly static	st
8	HXC-108_tui_views/helixcode-tui-qa-...mp4	50	5.00	10	3 (6%)	DIFFERENT	mostly static	st
9	HXC-108_tui_views/helixcode-tui-sessions-...mp4	50	5.00	10	3 (6%)	DIFFERENT	mostly static	st
10	HXC-108_tui_views/helixcode-tui-tasks-...mp4	50	5.00	10	3 (6%)	DIFFERENT	mostly static	st
11	HXC-108_tui_views/helixcode-tui-workers-...mp4	50	5.00	10	3 (6%)	DIFFERENT	mostly static	st
12	HXC-108_web/helixcode-web-llmchat-...mp4	419	13.97	30	8 (2%)	DIFFERENT	streaming	liv ~ @0 —

iOS: no committed mp4 — evidence is a single frame + ffprobe stamp (per the run notes).
Honestly noted: liveness oracle N/A (no video to analyze). Not counted in the 12; not a FROZEN-BLUFF (it never claimed a video).

Interpretation (§11.4.6)

- **Zero FROZEN-BLUFF findings.** No recording is a single frozen/stale image. The decisive proof: **every one of the 12 has first-frame $\neq$ last-frame AND ≥ 3 distinct (mpdecimate-survivor) frames** — a frozen-stale bluff would show exactly 1 distinct frame and identical first/last.
 - The **6 “live”** recordings (android, 3× cli, server-api, dashboard, web-llmchat) show clear content advancement: the Android client at 76% distinct frames is unambiguously live; the dashboard updates periodically (0/0.1/5.1/10.1s); the web LLM-chat advances every ~2s consistent with token streaming.
 - The **5 TUI-view “static-real”** recordings (projects/qa/sessions/tasks/workers) legitimately render the view once then sit — 3 distinct frames (initial blank → rendered view → minor change) over a 5s window. This is *rendered-but-static real UI*, NOT a frozen black/stale frame: each has a genuine content transition (first $\neq$ last) and the OCR-content-verification (prior run) already confirmed the right content is on screen. Per §11.4.107 honest boundary, a deliberately-static real screen with a real distinct frame count is acceptable and **MUST NOT** be false-FAILED as frozen.
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Overall verdict

PASS — liveness confirmed for all 12 committed durable mp4s. 6 genuinely-advancing (live), 5 static-but-real (real content, expected few changes for a TUI status/list screen), 1 (android) strongly live. **No frozen-stale-bluff detected; no re-record required.** iOS honestly carries no committed mp4 (frame + ffprobe-stamp only) — liveness oracle not applicable there.

This confirmation **COMPLEMENTS** (does not replace) the prior OCR content-verification and the §11.4.5 captured-evidence layer.